

Blood & Bone Marrow Histopathology - UPDATED Factoids

Released Schoenlein SAL review slides integrated • exact-slide cues first

Deck sequence: neutrophil -> lymphocyte -> monocyte -> eosinophil -> basophil -> platelets -> peripheral blood -> bone marrow -> megakaryocytes.

Five WBCs - practical identification

Cell	What to see	Released-slide wording / association	Fast discriminator
Neutrophil	Lobulated nucleus	Most abundant granulocyte; phagocytosis	The segmented/lobulated one
Lymphocyte	Spherical nucleus; very little cytoplasm; agranular	Cannot distinguish T vs B without special stain	Nucleus nearly fills cell
Monocyte	Large cell; horseshoe-shaped nucleus; agranular	Becomes a phagocytic cell after entering connective tissue	Biggest-looking agranular cell
Eosinophil	Bright eosinophilic cytoplasmic granules	Allergic reactions, parasitic infections, phagocytosis	Bright pink/red granules
Basophil	Large basophilic cytoplasmic granules	Least numerous granulocyte	Dark purple granules dominate

Granulocytes in the released deck

Neutrophil • eosinophil • basophil. The nucleus/granule pattern is the main visual discriminator.

Agranular in the released deck

Lymphocyte • monocyte. Lymphocyte = spherical/scant cytoplasm; monocyte = large/horseshoe nucleus.

Platelets - Slide 7

Small fragmented elements in peripheral blood. They are fragments of megakaryocytes found in bone marrow and have important functions in blood clotting and wound healing.

Peripheral blood - Slide 8

Whole-field gestalt: erythrocytes dominate the field; leukocytes are scattered; platelets are tiny. Learn the overall “open” look before zooming in on any one cell.

Self-check from the released images

- Can you find a leukocyte before looking for its label?
- Can you identify the tiny platelet fragments without confusing them with stain debris?
- Can you tell lymphocyte from monocyte using nucleus shape and amount of cytoplasm?

Marrow + Official-LO RBC Supplement

Bone marrow - Slides 9-10

Peripheral blood	Bone marrow
Released slide labels erythrocytes, leukocytes, platelets	Denser and darker than peripheral blood
Mostly mature circulating elements	Contains precursor cells
Platelets are tiny fragments	Megakaryocytes are the platelet precursors
Relatively open smear field	Dense cellular tissue around bony elements / marrow spaces in the released image

Megakaryocyte = marrow giveaway. The released slide describes megakaryocytes as **large, multinucleated cells** and identifies them as platelet precursors.

Official-LO RBC abnormalities NOT pictured in the 10-slide review deck

Finding	Recognition cue from the earlier course summary	Do not confuse with...
Normal RBC	Uniform anucleate biconcave disc; central pallor about 1/3 diameter	-
Anisocytosis	Variation in RBC size across the field	Poikilocytosis = shape variation
Microcytosis	RBCs smaller than expected, often with more central pallor if hypochromic	Spherocyte: small/dense but lacks pallor
Spherocytosis	Dense round cells with little or no central pallor	Typical microcyte with retained pallor
Sickled cells	Crescent / elongated pointed cells	Normal disc-shaped RBC
Reticulocytosis	Polychromasia: slightly larger bluish-gray RBCs from residual RNA	Nucleated marrow precursor

What the released deck DOES cover
5 WBCs • platelets • normal peripheral field • marrow field • megakaryocytes.

What it DOES NOT picture
Dedicated RBC-abnormality panels • staged marrow aspirate maturation sequences.

Last 30 seconds: lobulated neutro • spherical/scant lymphocyte • horseshoe monocyte • bright eos granules • dark baso granules • tiny platelet fragments • marrow = denser/darker + precursors + megakaryocytes.

Primary source: released Schoenlein Blood and Bone Marrow Histopathology review deck. RBC abnormality table is explicitly a prior-summary / official-LO supplement.